RAMAIAH
Institute of Technology

Dept. of Computer Science and Engineering (AI & ML) & Computer Science and Engineering (Cyber Security)

Programme: B E – Computer Science and Engineering (AI&ML) & Computer Science and Engineering (Cyber Security)

Internal Assessment – I

TERM : 04-10-2023 to 27-01-2024	COURSE NAME: ACN
DATE : 28-11-2023 TIME: 03.30 PM – 04.30 PM	COURSE CODE : CIE552
MAX MARKS: **30**	PORTIONS : 2 Units

Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.** Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Explain the three strategies involved in transition from IPv4 protocol to IPv6 protocol.	L1	CO1	5M
b	Given the IPV4 addresses 205.16.37.39/28 and 25.34.12.56/16. What are the first and last addresses of blocks.	L2	CO1	5M
c	Describe the detailed structure and enumerate the various fields of an IPV4 packet.	L2	CO1	5M
2.a	Explain the concept of ARP and how it facilitates communication between devices in a network. Provide a scenario where a lack of ARP understanding might lead to connectivity issues.	L1	CO2	5M
b	Enumerate the specific routing metrics commonly used in distance vector protocols and explain how these metrics influence the decision-making process of routers in updating their routing tables.	L2	CO2	5M
c	Deliberate a scenario: Network Diagnostics at Cloud Innovations Inc. Background: Cloud Innovations Inc. has been experiencing intermittent connectivity issues between their on-premises servers and cloud-based resources. As the network administrator, you decide to leverage ICMP for diagnostic purposes. Describe the specific ICMP error messages you would expect to encounter in this scenario.	L3	CO2	5M
3.a	An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to two groups of customers as follows: i. The first group has 64 customers; each needs 256 addresses ii. The second group has 128 customers; each needs 128 addresses Design the subblocks and find out how many addresses are still available after these allocations.	L3	CO1	5M

b	Illustrate the various techniques involved in forwarding a packet from source to the destination.	L3	CO2	5M
c	Differentiate direct and indirect delivery in network communication	L2	CO2	5M

* **L1** – Remember, **L2** – Understand, **L3**- Apply, **L4**- Analyze, **L5**-Evaluate, **L6**-Create